

Multivariable Calculus Notes

Hemanta Mandal

1 Neighbourhood, Bounded Functions, and Limits

Neighbourhood of a Point

The δ -**neighbourhood** of a point $P(x_0, y_0)$ is the set of all points which lie inside a circle of radius δ with center at P . It is denoted by $N_\delta(P)$ or $N(P; \delta)$:

$$N_\delta(P) = \left\{ (x, y) \mid \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta \right\}$$

Bounded Function

A function $f(x, y)$ is said to be **bounded** in some domain D if there exists a real finite positive number M such that:

$$|f(x, y)| \leq M \quad \text{for all } (x, y) \in D$$

Limit of a Function of Two Variables

Let $z = f(x, y)$ be a function defined in a domain D . Let $P(x_0, y_0)$ be a point in D . If for a given real number $\epsilon > 0$, there exists a $\delta > 0$ such that for every point (x, y) in the δ -neighbourhood of $P(x_0, y_0)$:

$$|f(x, y) - L| < \epsilon \quad \text{whenever} \quad \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$$

then we write:

$$\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = L$$

2 Continuity of Functions of Two Variables

A function $f(x, y)$ is said to be **continuous** at (x_0, y_0) if:

1. $f(x_0, y_0)$ is defined.
2. $\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y)$ exists.
3. $\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = f(x_0, y_0)$.

In ϵ - δ definition: $f(x, y)$ is continuous at (x_0, y_0) if

$$|f(x, y) - f(x_0, y_0)| < \epsilon \quad \text{whenever} \quad \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$$

Examples: Limits and Continuity

Example 1 (Single Variable Demonstration): Prove that $\lim_{x \rightarrow 1} (2x + 1) = 3$.

Solution:

We need $|f(x) - L| < \epsilon \implies |(2x + 1) - 3| < \epsilon$:

$$|2x - 2| < \epsilon \implies 2|x - 1| < \epsilon \implies |x - 1| < \frac{\epsilon}{2}$$

Choosing $\delta = \frac{\epsilon}{2}$, for example if $\delta = 0.1$, then $\epsilon = 0.2$.

Example 2: Show using ϵ - δ definition that $\lim_{(x,y) \rightarrow (2,1)} (3x + 4y) = 10$.

Solution:

$$|f(x, y) - 10| = |3x + 4y - 10| = |3(x - 2) + 4(y - 1)| \leq 3|x - 2| + 4|y - 1|$$

Since $|x - 2| \leq \sqrt{(x - 2)^2 + (y - 1)^2} < \delta$ and $|y - 1| < \delta$:

$$|3(x - 2) + 4(y - 1)| < 3\delta + 4\delta = 7\delta$$

Setting $7\delta < \epsilon \implies \delta < \frac{\epsilon}{7}$. Thus the limit is 10.

Example 3 (Non-existence of Limit): Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$.

Solution:

Approach along paths $y = mx$:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x(mx)}{x^2 + (mx)^2} = \lim_{x \rightarrow 0} \frac{mx^2}{x^2(1 + m^2)} = \frac{m}{1 + m^2}$$

Since the limit depends on the slope m , different paths yield different limit values. Hence, the limit **does not exist**.

Example 4: Show that $f(x, y)$ is continuous at $(0, 0)$, where:

$$f(x, y) = \begin{cases} \frac{2x^4 + 3y^4}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

Solution:

Using polar coordinates $x = r \cos \theta$, $y = r \sin \theta$:

$$|f(x, y) - f(0, 0)| = \left| \frac{r^4(2 \cos^4 \theta + 3 \sin^4 \theta)}{r^2} \right| = r^2 |2 \cos^4 \theta + 3 \sin^4 \theta| \leq 5r^2$$

Since $r = \sqrt{x^2 + y^2} < \delta$:

$$5r^2 < 5\delta^2 < \epsilon \implies \delta = \sqrt{\frac{\epsilon}{5}}$$

Since $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0 = f(0, 0)$, the function is continuous at $(0, 0)$.

Example 5: Show that $f(x, y)$ is continuous at $(0, 0)$, where:

$$f(x, y) = \begin{cases} \frac{x^2 + y^2}{|x| + |y|}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

Solution:

Using the inequality $(|x| + |y|)^2 \leq 2(x^2 + y^2)$:

$$|f(x, y) - f(0, 0)| = \frac{x^2 + y^2}{|x| + |y|} \leq \frac{2(|x| + |y|)^2}{|x| + |y|} = 2(|x| + |y|) \leq 2\sqrt{x^2 + y^2} < 2\delta$$

Choosing $\delta = \frac{\epsilon}{2}$ proves continuity at $(0, 0)$.

3 Partial Derivatives

Definition

The derivative of a function of several variables with respect to one independent variable, keeping all other independent variables constant, is called the ****partial derivative****.

$$f_x = \frac{\partial f}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

$$f_y = \frac{\partial f}{\partial y} = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$

Theorem on Continuity via Partial Derivatives

A function f is continuous at (x_0, y_0) if both of its partial derivatives exist and are bounded in a neighbourhood of (x_0, y_0) .

$$f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0) = \Delta x f_x(x_0, y_0 + \Delta y) + \Delta y f_y(x_0, y_0) \rightarrow 0 \quad \text{as } \Delta x, \Delta y \rightarrow 0$$

Example: Show that $f(x, y) = (x + y) \sin\left(\frac{1}{x+y}\right)$ for $x + y \neq 0$ and $f(0, 0) = 0$ is continuous at $(0, 0)$, but its partial derivatives do not exist at $(0, 0)$.

Solution:

1. ****Continuity****

$$|f(x, y) - f(0, 0)| = \left| (x + y) \sin\left(\frac{1}{x+y}\right) \right| \leq |x + y| \leq 2\sqrt{x^2 + y^2} < 2\delta < \epsilon$$

Choosing $\delta = \frac{\epsilon}{2}$, $f(x, y)$ is continuous at $(0, 0)$.

2. ****Partial Derivative at $(0, 0)$ ****

$$f_x(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - f(0, 0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\Delta x \sin\left(\frac{1}{\Delta x}\right)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \sin\left(\frac{1}{\Delta x}\right)$$

This limit ****does not exist**** due to oscillation. Hence $f_x(0, 0)$ and $f_y(0, 0)$ do not exist.

4 Differentiability and Linear Approximation

Single Variable Linearization Review

For $f(x)$, the linear approximation $L(x)$ at $x = a$ is:

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \implies L(x) = f(a) + f'(a)(x - a)$$

Two-Variable Linearization Definition

Let $h(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$. The function $f(x, y)$ is ****differentiable**** at (a, b) if:

$$\lim_{(x,y) \rightarrow (a,b)} \frac{f(x, y) - h(x, y)}{\|(x, y) - (a, b)\|} = 0$$

Example 1: Find the linear approximation $h(x, y)$ of $f(x, y) = xy + 2x + y$ at $(0, 0)$.

Solution:

$$f(0, 0) = 0$$

$$f_x = y + 2 \implies f_x(0, 0) = 2$$

$$f_y = x + 1 \implies f_y(0, 0) = 1$$

$$h(x, y) = 0 + 2x + y = \mathbf{2x + y}$$

Checking differentiability condition:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{(xy + 2x + y) - (2x + y)}{\sqrt{x^2 + y^2}} = \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2 + y^2}}$$

In polar coordinates ($x = r \cos \theta, y = r \sin \theta$):

$$\lim_{r \rightarrow 0} \frac{r^2 \cos \theta \sin \theta}{r} = \lim_{r \rightarrow 0} r \cos \theta \sin \theta = 0$$

Thus, $f(x, y)$ is differentiable at $(0, 0)$.

Example 2: Show that $f(x, y) = \begin{cases} \frac{x^3 + 2y^3}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$ is ****not differentiable**** at $(0, 0)$.

Solution:

Partial derivatives at $(0, 0)$:

$$f_x(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - f(0, 0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{(\Delta x)^3 / (\Delta x)^2}{\Delta x} = 1$$

$$f_y(0, 0) = \lim_{\Delta y \rightarrow 0} \frac{f(0, \Delta y) - f(0, 0)}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{2(\Delta y)^3 / (\Delta y)^2}{\Delta y} = 2$$

Linear approximation: $h(x, y) = x + 2y$.

Testing limit condition:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\frac{x^3+2y^3}{x^2+y^2} - (x+2y)}{\sqrt{x^2+y^2}} = \lim_{(x,y) \rightarrow (0,0)} \frac{-xy^2 - 2x^2y}{(x^2+y^2)^{3/2}}$$

In polar coordinates:

$$= -\cos \theta \sin \theta (\sin \theta + 2 \cos \theta)$$

This limit depends on θ and does not tend to zero. Therefore, $f(x, y)$ is ****not differentiable**** at $(0, 0)$.

5 Taylor's Theorem for Two Variables

Taylor series expansion of $f(x, y)$ around (a, b) :

$$\begin{aligned} f(x, y) &= f(a, b) + [f_x(a, b)(x - a) + f_y(a, b)(y - b)] \\ &\quad + \frac{1}{2!} [f_{xx}(a, b)(x - a)^2 + 2f_{xy}(a, b)(x - a)(y - b) + f_{yy}(a, b)(y - b)^2] + \dots \end{aligned}$$

Or using increments $h = x - x_0$, $k = y - y_0$:

$$f(x_0 + h, y_0 + k) = f(x_0, y_0) + \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}\right) f(x_0, y_0) + \frac{1}{2!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}\right)^2 f(x_0, y_0) + \dots$$

6 Taylor Polynomials of Two Variables

Problem: Determine the 2nd degree Taylor polynomial for

$$f(x, y) = \sin 2x + \cos y \quad \text{at } (0, 0)$$

First-Order Partial Derivatives and Values at $(0, 0)$:

$$f(x, y) = \sin 2x + \cos y \implies f(0, 0) = \sin 0 + \cos 0 = 1$$

$$f_x = 2 \cos 2x \implies f_x(0, 0) = 2 \cos 0 = 2$$

$$f_y = -\sin y \implies f_y(0, 0) = -\sin 0 = 0$$

Linear Approximation:

$$L(x, y) = f(0, 0) + f_x(0, 0)(x - 0) + f_y(0, 0)(y - 0) = 1 + 2x$$

Second-Order Partial Derivatives and Values at $(0, 0)$:

$$f_{xx} = -4 \sin 2x \implies f_{xx}(0, 0) = -4 \sin 0 = 0$$

$$f_{xy} = 0 \implies f_{xy}(0, 0) = 0$$

$$f_{yy} = -\cos y \implies f_{yy}(0, 0) = -\cos 0 = -1$$

Second-Degree Taylor Polynomial $Q(x, y)$:

$$\begin{aligned} Q(x, y) &= 1 + 2x + \frac{f_{xx}(0, 0)}{2}(x - 0)^2 + f_{xy}(0, 0)(x - 0)(y - 0) + \frac{f_{yy}(0, 0)}{2}(y - 0)^2 \\ &= 1 + 2x + \frac{0}{2}x^2 + 0xy + \left(-\frac{1}{2}\right)y^2 \\ &= 1 + 2x - \frac{y^2}{2} \end{aligned}$$

—

7 Error Analysis for Taylor Series

Error Bound Formula

The error E_L in linear approximation is bounded by:

$$E_L = \frac{1}{2!} [(x - x_0)^2 f_{xx} + 2(x - x_0)(y - y_0)f_{xy} + (y - y_0)^2 f_{yy}]$$

Let $B_L = \max\{|f_{xx}|, |f_{xy}|, |f_{yy}|\}$. Then:

$$|E_L| \leq \frac{B_L}{2} [|x - x_0|^2 + 2|x - x_0||y - y_0| + |y - y_0|^2] = \frac{B_L}{2} (|x - x_0| + |y - y_0|)^2$$

—

Example 1: Find the linear Taylor series for $f(x, y) = 2x^3 + 3y^3 - 4x^2y$ at $(1, 2)$, and find the error bound in the region $|x - 1| < 0.01$ and $|y - 2| < 0.1$.

Solution:

$$|E_L| \leq \frac{B_L}{2} (|x - 1| + |y - 2|)^2 \leq \frac{B_L}{2} (0.01 + 0.1)^2 = 0.00605 B_L$$

Bounding second derivatives:

$$\begin{aligned} \max |f_{xx}| &= \max |12x - 8y| = \max |12(x - 1) - 8(y - 2) + 4| \\ &\leq 12 \times 0.01 + 8 \times 0.1 + 4 = 4.92 \\ |f_{xy}| &\leq 8.08, \quad |f_{yy}| \leq 37.8 \end{aligned}$$

Thus, $B_L = 37.8$, and:

$$E_L \leq 0.00605 \times 37.8 \approx 0.23$$

—

Example 2: Consider $f(x, y) = xe^y + 1$ around $(1, 0)$.

Partial Derivatives:

$$\begin{aligned} f_x &= e^y, \quad f_y = xe^y \\ f(1, 0) &= 1 \cdot e^0 + 1 = 2 \\ f_x(1, 0) &= e^0 = 1, \quad f_y(1, 0) = 1 \cdot e^0 = 1 \end{aligned}$$

Linear approximation:

$$L(x, y) = f(1, 0) + f_x(1, 0)(x - 1) + f_y(1, 0)(y - 0) = 2 + (x - 1) + y = 1 + x + y$$

Second Derivatives at (1, 0):

$$\begin{aligned} f_{xx} &= 0 \implies f_{xx}(1, 0) = 0 \\ f_{xy} &= e^y \implies f_{xy}(1, 0) = e^0 = 1 \\ f_{yy} &= xe^y \implies f_{yy}(1, 0) = 1 \cdot e^0 = 1 \end{aligned}$$

Second-Degree Taylor Polynomial:

$$\begin{aligned} Q(x, y) &= 1 + x + y + \frac{1}{2} [0 \cdot (x - 1)^2 + 2(1)(x - 1)(y - 0) + 1 \cdot (y - 0)^2] \\ &= 1 + x + y + \frac{1}{2}(2xy - 2y + y^2) \\ &= 1 + x + xy + \frac{y^2}{2} \end{aligned}$$

—

8 Euler's Theorem for Homogeneous Functions

Definition

A function $f(x, y)$ is homogeneous of degree n if $f(\lambda x, \lambda y) = \lambda^n f(x, y)$.

Euler's Theorem (First Order)

If $f(x, y)$ is a homogeneous function of degree n , then:

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = nf$$

Proof Outline: Let $f(x, y) = x^n g(y/x)$. Differentiating with respect to x and y :

$$\begin{aligned} \frac{\partial f}{\partial x} &= nx^{n-1}g(y/x) - yx^{n-2}g'(y/x) \\ \frac{\partial f}{\partial y} &= x^{n-1}g'(y/x) \end{aligned}$$

Multiplying and adding:

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = nx^n g(y/x) = nf$$

—

Euler's Theorem (Second Order)

$$x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} = n(n-1)f$$

—

Example 1: For $u = x^2 \tan^{-1}(y/x) - y^2 \tan^{-1}(x/y)$, show that:

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 2(2-1)u = 2u$$

Example 2: Let $u(x, y) = \cos^{-1} \left(\frac{x+y}{\sqrt{x}+\sqrt{y}} \right)$.

Let $\cos u = \frac{x+y}{\sqrt{x}+\sqrt{y}}$, which is homogeneous of degree 1/2. Applying Euler's theorem to $f = \cos u$:

$$\begin{aligned} x \frac{\partial}{\partial x}(\cos u) + y \frac{\partial}{\partial y}(\cos u) &= \frac{1}{2} \cos u \\ -\sin u \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right) &= \frac{1}{2} \cos u \implies x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = -\frac{1}{2} \cot u \end{aligned}$$

9 Extrema of Functions of Two Variables

Definitions

- A point (a, b) is a **local minimum** if $f(a+h, b+k) \geq f(a, b)$ for all small h, k .
- A point (a, b) is a **local maximum** if $f(a+h, b+k) \leq f(a, b)$ for all small h, k .

Second Derivative Test

At a critical point (a, b) where $f_x(a, b) = 0$ and $f_y(a, b) = 0$, let:

$$A = f_{xx}, \quad B = f_{xy}, \quad C = f_{yy}$$

The quadratic form approximation gives:

$$Ah^2 + 2Bhk + Ck^2 = A \left(h + \frac{B}{A}k \right)^2 + \frac{AC - B^2}{A}k^2$$

1. If $AC - B^2 > 0$ and $A > 0 \implies$ Local Minimum.
 2. If $AC - B^2 > 0$ and $A < 0 \implies$ Local Maximum.
 3. If $AC - B^2 < 0 \implies$ Saddle Point.
 4. If $AC - B^2 = 0 \implies$ Test Inconclusive.
-

Example: Find the absolute maximum and minimum of $f(x, y) = 4x^2 + 9y^2 - 8x - 12y + 4$ over $x = 2, y = 3$ and coordinate axes.

1. Interior Critical Points:

$$f_x = 8x - 8 = 0 \implies x = 1, \quad f_y = 18y - 12 = 0 \implies y = 2/3$$

Critical point: $(1, 2/3)$.

$$r = f_{xx} = 8, \quad s = f_{xy} = 0, \quad t = f_{yy} = 18$$

$$rt - s^2 = 144 > 0 \quad \text{and} \quad r > 0 \implies \text{Minimum at } (1, 2/3), \quad f(1, 2/3) = -4$$

2. Boundary Analysis:

- On $y = 0$: $g(x) = 4x^2 - 8x + 4 \implies g'(x) = 8x - 8 = 0 \implies x = 1$.
 $f(0, 0) = 4, f(2, 0) = 4$.
- On $x = 2$: $g(y) = 9y^2 - 12y + 4 \implies g'(y) = 18y - 12 = 0 \implies y = 2/3$.
 $f(2, 2/3) = 0, f(2, 3) = 49$.
- On $y = 3$: $g(x) = 4x^2 - 8x + 49 \implies x = 1$. $f(1, 3) = 45, f(0, 3) = 49$.
- On $x = 0$: $g(y) = 9y^2 - 12y + 4 \implies y = 2/3$.

Conclusion:

$$\text{Minimum Value} = -4 \quad \text{at } (1, 2/3)$$

$$\text{Maximum Value} = 49 \quad \text{at } (2, 3) \text{ and } (0, 3)$$

10 Beta and Gamma Functions

Definitions

Gamma Function:

$$\Gamma(p) = \int_0^\infty x^{p-1} e^{-x} dx \quad (p > 0)$$

Beta Function:

$$B(p, q) = \int_0^1 x^{p-1} (1-x)^{q-1} dx \quad (p > 0, q > 0)$$

Symmetry Property of Beta Function

$$B(p, q) = B(q, p)$$

Proof: Substitute $x = 1 - y \implies dx = -dy$:

$$B(p, q) = \int_1^0 (1-y)^{p-1} y^{q-1} (-dy) = \int_0^1 y^{q-1} (1-y)^{p-1} dy = B(q, p)$$

Example 1: Evaluate $\int_0^1 x^m(1-x^2)^n dx$ for $m > -1, n > -1$.

Solution: Substitute $x^2 = y \implies 2x dx = dy \implies x dx = \frac{1}{2} dy$:

$$\begin{aligned}\int_0^1 x^m(1-x^2)^n dx &= \int_0^1 x^{m-1}(1-x^2)^n(x dx) \\ &= \frac{1}{2} \int_0^1 y^{\frac{m-1}{2}}(1-y)^n dy \\ &= \frac{1}{2} B\left(\frac{m+1}{2}, n+1\right)\end{aligned}$$

Example 2: Evaluate $\int_a^b (x-a)^{m-1}(b-x)^{n-1} dx$.

Solution: Let $x = a + (b-a)y \implies dx = (b-a) dy$. When $x = a \implies y = 0$; when $x = b \implies y = 1$:

$$\begin{aligned}I &= \int_0^1 [(b-a)y]^{m-1}[(b-a)(1-y)]^{n-1}(b-a) dy \\ &= (b-a)^{m+n-1} \int_0^1 y^{m-1}(1-y)^{n-1} dy \\ &= (b-a)^{m+n-1} B(m, n)\end{aligned}$$

Example 3 (Trigonometric Form of Beta Function):

$$\int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta = \frac{1}{2} B\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$$

Proof:

$$\int_0^{\pi/2} \sin^{p-1} \theta \cos^{q-1} \theta (\sin \theta \cos \theta) d\theta$$

Substitute $x = \sin^2 \theta \implies dx = 2 \sin \theta \cos \theta d\theta$:

$$\begin{aligned}I &= \int_0^1 (x^{\frac{1}{2}})^{p-1} (1-x)^{\frac{q-1}{2}} \frac{1}{2} dx \\ &= \frac{1}{2} \int_0^1 x^{\frac{p+1}{2}-1} (1-x)^{\frac{q+1}{2}-1} dx \\ &= \frac{1}{2} B\left(\frac{p+1}{2}, \frac{q+1}{2}\right)\end{aligned}$$

Example 4 (Alternative Integral Form of Beta Function): Show that:

$$B(m, n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx \quad (m > 0, n > 0)$$

Proof: Start with $B(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1} dx$. Substitute $x = \frac{1}{1+y} \implies dx = -\frac{1}{(1+y)^2} dy$. When $x \rightarrow 0 \implies y \rightarrow \infty$; when $x \rightarrow 1 \implies y \rightarrow 0$:

$$\begin{aligned} B(m, n) &= \int_{\infty}^0 \left(\frac{1}{1+y} \right)^{m-1} \left(1 - \frac{1}{1+y} \right)^{n-1} \left(-\frac{1}{(1+y)^2} \right) dy \\ &= \int_0^{\infty} \frac{y^{n-1}}{(1+y)^{m+n}} dy \end{aligned}$$

By dummy variable replacement $y \rightarrow x$:

$$B(m, n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

11 Gamma Function Properties

Definition:

$$\Gamma(p) = \int_0^{\infty} x^{p-1} e^{-x} dx$$

Properties and Proofs

1. $\Gamma(1) = 1$

$$\Gamma(1) = \int_0^{\infty} e^{-x} dx = -e^{-x} \Big|_0^{\infty} = -[0 - 1] = 1$$

2. $\Gamma(p+1) = p\Gamma(p)$ Using integration by parts:

$$\begin{aligned} \Gamma(p+1) &= \int_0^{\infty} e^{-x} x^p dx \\ &= \left[x^p \int e^{-x} dx - \int \left(\frac{d}{dx} x^p \cdot \int e^{-x} dx \right) \right]_0^{\infty} \\ &= [-x^p e^{-x}]_0^{\infty} + p \int_0^{\infty} e^{-x} x^{p-1} dx \\ &= 0 + p\Gamma(p) = p\Gamma(p) \end{aligned}$$

3. If $p = n$ is an integer:

$$\Gamma(n+1) = n\Gamma(n) = n(n-1)\Gamma(n-1) = n(n-1)(n-2)\cdots 3 \cdot 2 \cdot 1 \cdot \Gamma(1) = n!$$

—

12 Relation Between Beta and Gamma Functions

4. **Proof Statement:**

$$B(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$$

Proof: Let $x = u^2$ and $y = v^2$:

$$\begin{aligned}\Gamma(m) &= \int_0^\infty e^{-u^2} (u^2)^{m-1} \cdot 2u \, du \\ \Gamma(n) &= \int_0^\infty e^{-v^2} (v^2)^{n-1} \cdot 2v \, dv\end{aligned}$$

Multiplying the two integrals gives:

$$\Gamma(m)\Gamma(n) = 4 \int_0^\infty \int_0^\infty e^{-(u^2+v^2)} u^{2m-1} v^{2n-1} \, du \, dv$$

Transforming to polar coordinates ($u = r \cos \theta, v = r \sin \theta$):

$$\begin{aligned}\frac{\partial u}{\partial r} &= \cos \theta, & \frac{\partial u}{\partial \theta} &= -r \sin \theta \\ \frac{\partial v}{\partial r} &= \sin \theta, & \frac{\partial v}{\partial \theta} &= r \cos \theta \\ |J| &= \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r \cos^2 \theta + r \sin^2 \theta = r\end{aligned}$$

Substituting into the integral:

$$\begin{aligned}\Gamma(m)\Gamma(n) &= 4 \int_0^{\pi/2} \int_0^\infty e^{-r^2} (r \sin \theta)^{2m-1} (r \cos \theta)^{2n-1} r \, dr \, d\theta \\ &= 4 \int_0^{\pi/2} \int_0^\infty r^{2m+2n-1} e^{-r^2} \sin^{2m-1} \theta \cos^{2n-1} \theta \, dr \, d\theta \\ &= 4 \left(\int_0^\infty r^{2m+2n-1} e^{-r^2} \, dr \right) \left(\int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta \, d\theta \right)\end{aligned}$$

Let $t = r^2 \implies dt = 2r \, dr$:

$$\begin{aligned}\int_0^\infty r^{2m+2n-1} e^{-r^2} \, dr &= \frac{1}{2} \int_0^\infty t^{m+n-1} e^{-t} \, dt = \frac{1}{2} \Gamma(m+n) \\ \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta \, d\theta &= \frac{1}{2} B\left(\frac{(2m-1)+1}{2}, \frac{(2n-1)+1}{2}\right) = \frac{1}{2} B(m, n)\end{aligned}$$

Substituting back:

$$\begin{aligned}\Gamma(m)\Gamma(n) &= 4 \left[\frac{1}{2} \Gamma(m+n) \right] \left[\frac{1}{2} B(m, n) \right] = \Gamma(m+n) B(m, n) \\ \implies B(m, n) &= \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}\end{aligned}$$

—

13 The Error Function (erf(x))

Definition:

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} \, dt$$

Properties:

1. $\operatorname{erf}(0) = 0$
2. $\operatorname{erf}(\infty) = \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-t^2} dt = \frac{2}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = 1$
3. Monotonically increasing function.
4. Odd function:

$$\operatorname{erf}(-x) = \frac{2}{\sqrt{\pi}} \int_0^{-x} e^{-t^2} dt$$

Let $t = -v \implies dt = -dv$:

$$\operatorname{erf}(-x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-(-v)^2} (-dv) = -\frac{2}{\sqrt{\pi}} \int_0^x e^{-v^2} dv = -\operatorname{erf}(x)$$

—

14 Multivariable Calculus Exercises

1. Find all the asymptotes of the curve:

$$x^3 - x^2y - xy^2 + y^3 + 2x^2 - 4y^2 + 2xy + x + y + 1 = 0$$

2. Show that the following function $f(x, y)$ is continuous at $(0, 0)$:

$$f(x, y) = \begin{cases} \frac{2x^3 + 3y^3}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

3. Find the Taylor series expansion of $f(x, y) = \tan^{-1}(x + 2y)$ at $(1, 0)$ up to 2nd order.

Answer: $\frac{1}{4}\pi - \frac{3}{4} + x + 2y - \frac{x^2}{4} - xy - y^2$